8 November 2026

Editors
Pattern Recognition Letters

Handedness for Bangla writer identification: Oracle gains and predicted-attribute utility

Dear Editors,

Please consider the enclosed manuscript as a research report in Pattern Recognition Letters.

The contribution is a controlled framework for testing whether predicted soft-biometric information improves identification. Attribute-classification accuracy and gains obtained with correct metadata do not establish the value of predicted attributes for identification. Our comparisons separate oracle opportunity, prediction errors, dependence between evidence sources, and writer-associated information. The resulting methodological finding is relevant to biometrics, document analysis and other recognition pipelines that combine a primary matcher with a predicted auxiliary attribute.

On BanglaHand, using 42 writers, six representations and 120 identification settings, known handedness improves top-1 accuracy by 6.1 percentage points; a known random binary partition provides a comparable 6.9-point benchmark. In contrast, calibrated fixed fusion changes accuracy by -0.6 points, properly nested learned weighting by -0.08 points, and hard filtering by -14.3 points. Neither fusion rule establishes a gain in the designated comparisons. Joint-correctness and decision-margin diagnostics explain why these results concern achieved identification utility rather than the predictability of handedness alone.

The distinct companion preprint, "Writer-independent handedness identification from offline Bangla handwriting under controlled acquisition" (doi:10.2139/ssrn.7662278), studies unseen-writer handedness prediction and photometric cues. The present manuscript reuses the same collection and overlapping representations; this relationship is explicitly cited and disclosed. Its question, identification endpoints, oracle and random-partition controls, nested fusion comparison, and error-dependence diagnostics are distinct from the companion study. We do not present the shared cohort as independent validation.

The manuscript is intended as a concise methodological study, with claims restricted to the tested collection and rules. It contains seven pages, with the 14-page supplement supplied separately. The supplementary reproducibility archive provides code, frozen features, numerical results and input hashes; the dataset is publicly available at doi:10.17632/n4sjstdzfy.1. Separate highlights and a graphical abstract accompany the manuscript.

Thank you for considering this manuscript.

Sincerely,

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